

Hence, the deuteron needs to capture an energy of $1879 - 1876 = 3 \text{ MeV}$

- 30 B** This shows the random nature of radioactive decay.
Option A: This is due to the law of decay, and not the spontaneous nature.
Option C: This is the spontaneous nature in which the rate of decay is unaffected by external conditions.
Option D: This is a fact not related to random nature.